MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIFTH SEMESTER MGU-UGP (HONOURS) / MGU-BBA (HONOURS) / MGU-BCA (HONOURS) REGULAR EXAMINATION OCTOBER 2026

Fifth Semester

Discipline Specific Core Course (DSC) - MG5DSCCSC301 - OPERATING SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Scheme for Evaluation

Part A: Very Short Answer Questions (Answer all) - (5 * 2 = 10 Marks)

1. **What is a Command Interpreter?** [U] [CO1]

   Answer: A command interpreter obtains and interprets the commands supplied by the user (1 mark) and executes them, either directly or by invoking the appropriate program (1 mark).

2. **Define response time in CPU scheduling.** [U] [CO2]

   Answer: Response time is the interval from the submission of a request (1 mark) until the first response begins; it does not include the time required to complete the response (1 mark).

3. **Explain how a binary semaphore provides mutual exclusion between two concurrent processes.** [U] [CO3]

   Answer: (1 mark each)

   - Initialise a binary semaphore to 1; each process executes an atomic wait() before entering the critical section, permitting only one process to enter (1 mark).
   - Execute signal() on exit from the critical section so that the other process can enter (1 mark).

4. **Identify the two major methods of recovering from deadlock.** [K] [CO3]

   Answer: (1 mark each)

   - Process termination (1 mark)
   - Resource preemption (1 mark)

5. **In linked allocation, if the first block of a file is known, how can the next block be located?** [A] [CO4]

   Answer: Read the pointer stored in the current file block (1 mark); this pointer identifies the next block in the file's linked sequence (1 mark).

Part B: Short Answer Questions (Answer any 6 out of 8 Questions) - (6 * 5 = 30 Marks)

6. **Outline the advantages and disadvantages of using a Command-Line Interface.** [U] [CO1]

   Answer: (1 mark for each point)

   - CLI accepts textual commands that are interpreted by a command interpreter (1 mark).
   - Advantage: permits direct and flexible specification of commands and options (1 mark).
   - Advantage: command files/scripts support automation of repeated operations (1 mark).
   - Disadvantage: requires users to learn command names, syntax and options (1 mark).
   - Disadvantage: commands are less discoverable; typing or syntax errors can lead to incorrect operations (1 mark).

7. **Compare batch operating system and time sharing operating system.** [K] [CO1]

   Answer: (1 mark for each correctly explained comparison; any five)

Basis	**Batch Operating System**	**Time-Sharing Operating System**
User interaction	Little or no interaction during execution of submitted jobs.	Interactive use during program execution.
Organisation	Jobs are submitted for batch processing.	CPU time is shared among active processes/users.
Execution	Processes jobs without requiring continuous user intervention.	Switches rapidly among processes to support interactive service.
Main emphasis	Efficient job processing and resource utilisation.	Short response time and interactive access.
Suitable workload	Jobs requiring little interaction.	Interactive programs and user sessions.

8. **Demonstrate how pipes can be used to transfer data between two related processes.** [U] [CO2]

   Answer: (1 mark for each point)

   - Explain a pipe as an IPC channel with a reading end and a writing end (1 mark).
   - Create the pipe before creating the child process; the child inherits access to it (1 mark).

- Sender closes its unused reading end; receiver closes its unused writing end (1 mark).
- Sender writes data into the pipe; receiver reads that data (1 mark).
- A coherent parent–child example or procedural demonstration, including closing the remaining ends after use (1 mark).

9. **Explain how the PCB is used during a context switch.** [A] [CO2]

   Answer: (1 mark for each point)

   - Identify the PCB as the structure holding process-management and execution-context information (1 mark).
   - Save the outgoing process's program counter, CPU registers and relevant processor state in its PCB (1 mark).
   - Update its process state and relevant scheduling information (1 mark).
   - Restore the incoming process's saved execution context from its PCB (1 mark).
   - Resume the incoming process from its saved position; identify context switching as overhead (1 mark).

10. **Explain the role of the producer, consumer, and bounded buffer in the Producer–Consumer Problem.** [U] [CO3]

    Answer: (1 mark for each point)

    - Producer generates items and places them in the buffer (1 mark).
    - Consumer removes items and processes/uses them (1 mark).
    - Bounded buffer has a fixed capacity and temporarily stores produced items (1 mark).
    - Producer waits when the buffer is full; consumer waits when it is empty (1 mark).
    - Synchronisation ensures mutual exclusion for shared buffer operations and prevents conflicting updates (1 mark).

11. **Summarize the role of hardware support in solving synchronization problems.** [U] [CO3]

    Answer: (1 mark for each point)

    - Need for atomic operations when accessing shared synchronisation variables (1 mark).
    - Disabling interrupts on a single processor and its limitations, particularly on multiprocessor systems (1 mark).
    - Atomic test-and-set instruction (1 mark).
    - Atomic swap/exchange (compare-and-swap) or another correctly explained equivalent hardware primitive (1 mark).
    - Using these primitives to implement locks and mutual exclusion; mention of busy waiting or other relevant limitations (1 mark).

12. **Compare resource preemption and resource ordering as methods of deadlock prevention.** [An] [CO3]

Answer: (1 mark for each point)

- Resource preemption prevents the no-preemption condition (1 mark).
- Releasing/preempting held resources when an additional request cannot be satisfied, with reacquisition before the process continues (1 mark).
- Applicable to resources whose state can be saved and restored (CPU registers, memory); limitations for other resources (printers, tape drives) (1 mark).
- Resource ordering prevents circular wait by assigning an order to resource types and requiring acquisition in increasing order (1 mark).
- Clear comparison: preemption removes held resources; ordering restricts the sequence of resource requests (1 mark).

13. **The available memory holes are 100 KB, 500 KB, 200 KB, 300 KB and 600 KB. Three processes of sizes 180 KB, 250 KB and 450 KB are to be allocated in the given order. Apply the first-fit, best-fit and worst-fit strategies and determine the memory block allocated to each process.** [A] [CO4]

Answer: Apply each strategy independently. Split the selected hole and retain its remainder for later allocations.

Strategy	**180 KB**	**250 KB**	**450 KB**
First-fit	500 KB hole → 320 KB remaining	320 KB remainder → 70 KB remaining	600 KB hole → 150 KB remaining
Best-fit	200 KB hole → 20 KB remaining	300 KB hole → 50 KB remaining	500 KB hole → 50 KB remaining
Worst-fit	600 KB hole → 420 KB remaining	500 KB hole → 250 KB remaining	Not allocated (largest remaining hole is 420 KB; process must wait)

Final free holes (in original address order):

Strategy	**Final free holes**
First-fit	100, 70, 200, 300, 150 KB
Best-fit	100, 50, 20, 50, 600 KB
Worst-fit	100, 250, 200, 300, 420 KB

Mark allocation: (1 mark for each point)

- Correct selection rules, including reuse of residual holes (1 mark).

- Correct first-fit allocations (1 mark).
- Correct best-fit allocations (1 mark).
- Correct worst-fit allocations, including failure to allocate 450 KB (1 mark).
- Correct intermediate working/remainders (1 mark).

Part C: Essay Questions (Answer any 2 out of 3 Questions) - (2 * 15 = 30 Marks)

14. **Explain the long-term, short-term and medium-term schedulers in detail. Discuss their functions, frequency of operation, scheduling decisions.** [U] [CO2]

Answer: (5 marks for each scheduler; 1 mark for each point)

Long-term scheduler (5 marks)

- Identified as the job/admission scheduler (1 mark).
- Selects jobs for admission into memory (1 mark).
- Controls the degree of multiprogramming (1 mark).
- Selects a suitable mix of CPU-bound and I/O-bound processes (1 mark).
- Operates relatively infrequently and makes admission decisions (1 mark).

Short-term scheduler (5 marks)

- Identified as the CPU scheduler (1 mark).
- Selects a process from the ready queue (1 mark).
- CPU-allocation decisions at relevant scheduling events (1 mark).
- Operates frequently and must be fast (1 mark).
- Distinguishes scheduler selection from the dispatcher's transfer of control (1 mark).

Medium-term scheduler (5 marks)

- Identifies the swapping/suspension role (1 mark).
- Temporarily removes processes from memory (1 mark).
- Subsequently reinstates (swaps in) processes (1 mark).
- Adjusts the degree of multiprogramming or relieves memory pressure (1 mark).
- Operates as required and makes suspension/resumption decisions (1 mark).

15. **a) Describe the data structures used by Banker's Algorithm and explain their purpose. b) Explain the steps followed by the Banker's Safety Algorithm to determine whether the system is safe.** [U] [CO3]

Answer: (a) 7 marks; (b) 8 marks

(a) Data structures and their purposes (7 marks)

- Define n processes and m resource types; identify appropriate vector/matrix dimensions (1 mark).
- Available: m-element vector giving currently available instances of each resource type (1 mark).
- Max: n × m matrix giving each process's declared maximum demand (1 mark).

- Allocation: n × m matrix giving resources currently assigned to each process (1 mark).
- Need: n × m matrix giving each process's remaining possible demand (1 mark).
- Relationship: Need[i, j] = Max[i, j] − Allocation[i, j] (1 mark).
- Temporary Work and Finish structures used by the safety test (1 mark).

(b) Safety Algorithm (8 marks)

- Initialise Work = Available (1 mark).
- Initialise Finish[i] = false for every process (1 mark).
- Find an unfinished process i such that $Need_i \leq$ Work for every resource type (2 marks).
- Simulate completion: Work = Work + $Allocation_i$ and set Finish[i] = true (2 marks).
- Repeat until all processes finish or no eligible unfinished process remains (1 mark).
- If all Finish values are true, the state is safe; otherwise it is unsafe. An unsafe state is not necessarily already deadlocked (1 mark).

16. **A logical address in a segmentation system is given as 2400. The segment table contains the following entries: Segment 0 (Base 1000, Limit 500), Segment 1 (Base 4000, Limit 800), Segment 2 (Base 6000, Limit 1000), Segment 3 (Base 8000, Limit 600). Use the segment table to determine whether the given logical address is valid and, if valid, calculate its corresponding physical address. Explain the steps involved in the address translation.** [A] [CO4]

Answer: (3 marks for each point)

- A segmented logical address consists of a pair (s, d), where s is the segment number and d is the offset (3 marks).
- Identifies that the encoding convention is not given and explains the ambiguity of the address 2400 (3 marks).
- Segment-table lookup: s indexes the table; Base gives the starting physical address and Limit gives the segment length (3 marks).
- Validity checks: $0 \leq s < 4$ and $0 \leq d <$ Limit[s]; an invalid segment number or an offset beyond the limit causes an addressing/protection trap (3 marks).
- Translation rule: Physical address = Base[s] + d, with the conclusion: no unique numerical result as printed; alternatively, 6400 under an explicitly stated segment-2/offset-400 assumption (3 marks).

Conditional numerical answer: If the candidate explicitly assumes that 2400 represents segment 2, offset 400:

- Segment 2 has Base 6000 and Limit 1000.
- Offset 400 < 1000, hence the address is valid.
- Physical address = 6000 + 400 = 6400.